Published by :
http://www.ijert.org

International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 11 Issue 02, February-2022

Contingency Analysis to Evaluate Power System Security and Voltage Stability for The Critical Buses

Using PowerWorld Simulator

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***Abstract*—This paper presents different contingency ranking and screening methods using linear sensitivity factors for the faster response of the system operators to estimate the post- contingency active power flow for all the system transmission lines. The study depends on the Powerworld simulator to investigate the single contingency scenario and rank the most critical lines using different formulas of performance indices. Then, by applying the multiple contingency scenarios using the bus outages, an investigation was performed using the complex power performance index to specify the critical system substations, that could happen because of any such missile attack on the critical substation or for any natural reasons such as earthquakes, hurricanes, floods, etc.**

Finally evaluate the critical buses voltage stability using PV and QV curves and assess the system stability under the highest PI contingences

Keywords*— *Contingency ranking, contingency screening, linear sensitivities, multiple contingency analysis, performance index, voltage stability, PV curves, QV curves

I. INTRODUCTION

System security entails methods aimed to keep the system running when components fail. The most typical failures in power systems are transmission lines and generating units. Transmission line failures affect power flows and bus voltages on the other lines. When a generator goes out of service in the power system, the operational conditions of the transmission lines and other generators, such as the system frequency, active and reactive power flow. [1] [2]

Planned outages as well as forced outages of transmission lines and generating units change the topology of the power system. As such, their impacts on the power system should be analyzed and rank the severity to take the proper action to maintain system reliability and keep customer satisfaction. [3] [4]

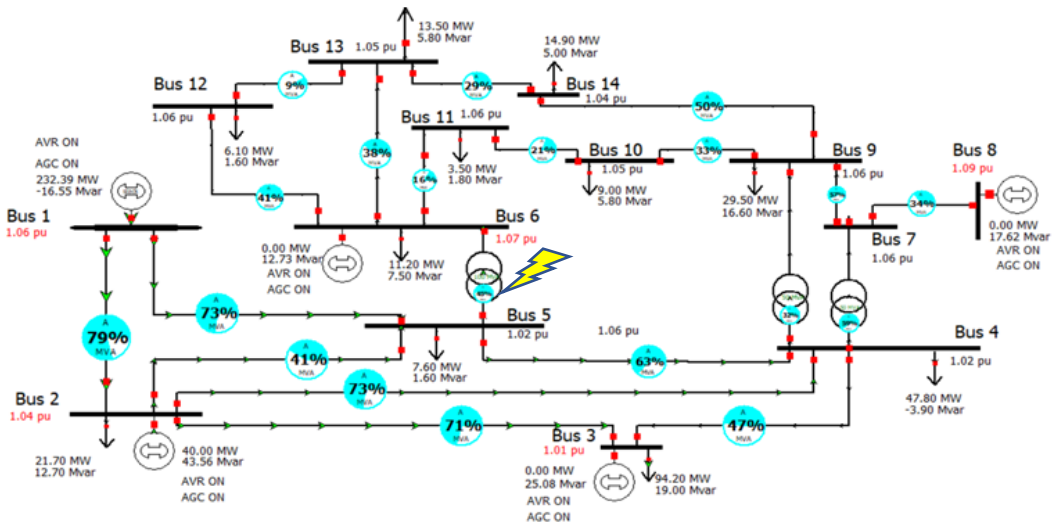


FIGURE 1. IEEE 14 bus system Single line diagram (AC-Simulator)

The main concern of power security is to sustain the flow of energy from the producing power plants to the loads centre (Local customers). However, the design of a power system should meet a certain degree of reliability, adequacy,

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International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 11 Issue 02, February-2022

quality, and security. to overcome the dynamic interruptions which might be distributed minor failures or major localized interruptions. [5]

Power systems that operate with a single outage or failure scenario are usually considered secure. If such a circumstance occurs, the widely used term "N-1" indicates that the system will stay in a secure working condition. So we need a fast methodology like sensitivity factors to examine the single element outages to rank and screen the less severe failures to evaluate the remaining using AC power flow or applying multiple outage "N-K". [6] [7]

The new advanced software such as Powerworld provides a wide range of functions for power flow, security analysis, system stability, and other complete packages that can help researchers in their work, as illustrated in figure (1).

After investigating the IEEE -14 bus using the Powerworld simulator, we find that line 10 (bus5-bus6) is the most vulnerable line by applying the N-1 scenario, and the substation represented by Bus2 is the most critical node in the system from the voltage stability point of view.

II. CONTINGENCY ANALYSIS FOR POWER SYSTEMS

Power system reliability takes priority in power system operations, particularly in large interconnected computerized power systems where widespread blackouts are a possibility. A power system should design to meet reliability requirements and operate economically. [8]

Power systems are huge, interconnected components that might collapse due to internal or external reasons, such as a circuit breaker malfunction or a severe weather situation. Transmission lines and generating units are the most common failures in power systems. Transmission line failures affect power flows and bus voltages on the other lines.

A. Contingency analysis Procedures:

Contingency analysis examines which transmission line or generator failure will result in a violation in line flows or bus voltages. To estimate system states, contingency analysis analyzes single and multiple outages. In the contingency analysis, the line flows and bus voltages are compared to their limitations.

Contingency screening, also known as contingency selection, is a technique in which the critical contingencies are picked using a DC power flow, and then the chosen contingencies are completely analyzed using an AC power flow. [9]

1) Linear Sensitivity Factors

Linear Sensitivity Factors "LSFs" are calculated using DC-Flow Analysis, which ignores bus voltages and MVAR flow and estimates the MW flow in each branch with a five percent accuracy. [10]To evaluate the critical cases by formulating a method to rank the contingency scenarios in such a manner that only those that are likely to result in limit violation need to be investigated in more depth, while the others are neglected. [11]

These LSFs represent the expected change in the transmission line flows due to a change in generation or network configuration. These variables can be obtained using many techniques, but they are mainly categorized into two categories:

- **Generation shift factors (GSF)**

The generator shift factors explain how the active power flow on transmission line l changes when the power generation unit at bus i disturbed as shown in Equation (1)

$$GSF_{l,i} = \frac{\Delta f_l}{\Delta P_i} = \frac{\Delta f_l}{-P_i^0} \tag{1}$$

Where, Δf_l represent the change in power flow of line l when the generation at bus i got out of service, $\Delta P_i = -P_i^0$ and the new power flow on each line in the network could be estimated as follows

$$f_l^1 = f_l^0 - GSF_{l,i}\ P_i^0 \tag{2}$$

The outage flow f_l^1 on each line can be compared to its limit f_l^{max} and those exceeding their limit announced as a violation. This would tell the operations personnel that the loss of the generator on bus i would result in an overload on line l. [11]

- **Line outage distribution factors (LODF)**

Line outage distribution factors (LODFs) calculate the line active power flow changes when the line outages take place in a power system, as shown in Equation (3)

$$LODF_{l,k} = \frac{\Delta f_l}{\Delta f_k} = \frac{\Delta f_l}{-f_k^0} \tag{3}$$

Where, Δf_l represent the change in power flow on line l when the line i got out of service, $\Delta f_k = -f_k^0$ and the post-Contingency power flow on line l with line k failure can be estimated as follows

$$f_l^1 = f_l^0 - LODF_{l,k}\, f_k^0 \tag{4}$$

For real-time security analysis, the impacts of hundreds or even thousands of outages on lines flows and bus voltages must be examined. This depicts the reality need for the speed, Accuracy, and acceptance of solution technique. In reality, solving with full AC power flow is not feasible for all contingency scenarios. Because the large number of these contingencies do not result a severe generation shifts or transmission lines violations. The objective is to eliminate the major part of harmless scenarios from the shortlist of branch contingencies. [10]

2) Performance Indices

The system performance index (PI) is a tool that could be used to estimate a contingency's relative severity. System

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International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 11 Issue 02, February-2022

performance indicators aren't all the same, and they take on various forms based on the criteria that concern most to the operator. However, physical elements of the system should be considered while choosing a performance index type. [8]

The most commonly type of system performance indices used as a measure of how much system variables like bus power injection or bus voltage differs from their rated values. For the PI, several relations have been proposed. The values we're interested in are used to choose which PI relationship to use. The following are the most common PI types:

- Active power based ranking methods

The active power performance index (PI_{MW}) gives a measure of line power flow overloads. It is given by the following equation:

$$PI_{MW} = \sum_{l=1}^{All\ Lines} w_l \left(\frac{P_l}{P_l^{lim}}\right)^2 \quad (5)$$

Where:

P_l Post- contingency Active power flow in line l

$P_l^{\lim}$ Active power capacity (limit) of line l

w_l Weighting factor of active power limit violation

This index ranks the contingencies that result in a large number of heavily loaded and overloaded lines in the system over those that result in a few lightly loaded lines.

- Reactive power or voltage security based ranking methods:

The Voltage Performance Index (PI_{QV}) is a technique used to measure bus violations and evaluate system inability resulting by out–of–limit bus voltage, it can be calculated as follows:

$$PI_{QV} = \sum_{i=1}^{all\ Buses} \left(w_i \frac{|V_i - V_i^{\lim}|}{V_i^{\lim}}\right) \quad (6)$$

Where:

V_i Post- contingency voltage at bus i

$V_i^{\lim}$ Bus voltage limit

w_i Weighting factor of voltage at bus i

This index ranks the contingencies that result in a large number of violated limit's bus voltage (could be overvoltage or undervoltage) in the system over those that didn't exceed the nominal bus voltage. [3]

III. : POWERWORLD SIMULATION FOR SINGLE CONTINGENCY N-1 SCENARIO

In this chapter, we will investigate the contingency analysis for the IEEE 14 bus system using the Powerworld program, The base voltage and power are considered as 138KV and 100MVA respectively.

TABLE 1. Bus data base-case AC solution

Bus Name	Voltage	Angle	Load		Generation	
	PU	Deg	MW	MVAR	MW	MVAR
Bus 1	1.06	0.00			232.39	-16.55
Bus 2	1.05	-4.98	21.70	12.70	40.00	43.56
Bus 3	1.01	-12.72	94.20	19.00	0.00	25.08
Bus 4	1.02	-10.31	47.80	-3.90		
Bus 5	1.02	-8.77	7.60	1.60		
Bus 6	1.07	-14.22	11.20	7.50	0.00	12.72
Bus 7	1.06	-13.36				
Bus 8	1.09	-13.36			0.00	17.62
Bus 9	1.06	-14.94	29.50	16.60		
Bus 10	1.05	-15.10	9.00	5.80		
Bus 11	1.06	-14.79	3.50	1.80		
Bus 12	1.06	-15.08	6.10	1.60		
Bus 13	1.05	-15.16	13.50	5.80		
Bus 14	1.04	-16.03	14.90	5.00		
			259.00	**73.50**	**272.39**	**82.43**

For the base-case situation the total generation is 285 MVA (272.4+j82.4) and the total load is 269 MVA (259+j73.5) and the spinning reserve in the system with losses the dynamic transient is 16 MVA which represent 7% of the largest generation unit located on Bus1.

TABLE 2. Lines data base-case AC solution

Line Number	Line Power	Line Capacity	Line Loading
	MVA	MVA	%
1	158.2	200	79.1
2	75.6	100	75.6
3	73.3	100	73.3
4	56.2	100	56.2
5	41.5	100	41.5
6	23.7	50	48.3
7	63.2	100	63.3
8	29.7	50	60.6
9	16.1	50	32.3
10	45.8	100	45.8
11	8.2	50	16.3
12	8.2	20	40.9
13	19.2	50	38.3
14	17.2	50	35.2
15	28.7	50	57.3
16	6.7	20	33.6
17	10.1	20	50.5
18	4.1	20	20.7
19	1.8	20	8.9
20	5.9	20	29.5

After built the system and obtain the base-case result for the lines and buses, and then perform the contingency analysis using DC and AC power flow analysis using Powerworld simulator under single line loess scenario, the simulator report that the DC power flow have a total run time to complete the contingency analysis is 0.21 sec compared to 0.35 Sec for AC power flow as shown in figure (1) below

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International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 11 Issue 02, February-2022

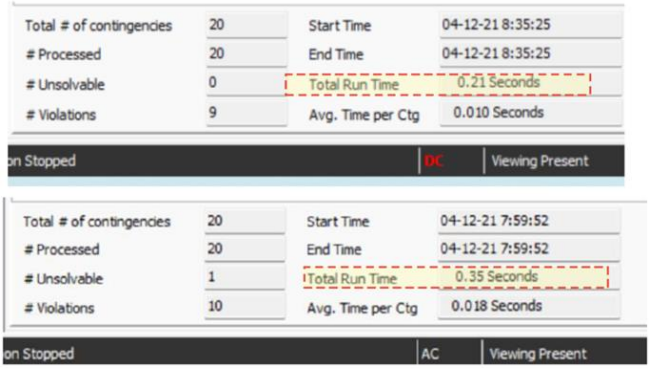


FIGURE 2. DC and AC contingency simulator summary

A. *N-1 DC Contingency Analysis Results:*

By applying N-1 contingency analysis using the DC power flow analysis as illustrated in (Appendix-1), the performance indices calculated separately for each case as shown below in table (3)

As pert the result form DC power flow, Line 10 had ranked as the highest power flow performance index indicating that this line is the most critical contingency in the system with 5 overloads line and largest (PI=12.17), then Line 1 outage which overloads 2, then Line 3 outage with one overload and last Line2 with one overload also.

TABLE 3. N-1 contingency ranking using DC power flow

Ranking	Violations	PI_{DC}
Line 10	5	12.17
Line 1	2	9.46
Line 3	1	8.48
Line 2	1	4.96

B. *N-1 AC Contingency Analysis Results:*

By applying N-1 contingency analysis using the AC power flow analysis as illustrated in (Appendix-2), the performance indices calculated separately for each case as shown below in table (4)

TABLE 4. N-1 contingency ranking using AC power flow

Ranking	Violations	PI_{AC}
Line 10	6	13.60
Line 3	2	9.64
Line 13	1	6.74
Line 2	1	5.75

Using AC power flow, again Line 10 has the highest power flow performance index indicating that this line is the most critical contingency in the system with 6 overloads line and largest (PI=13.6), then Line 1 outage which overloads 2, then Line 3 outage with one overload and last Line2 with one overload also.

As noticed the use of DC load flow models provides acceptable capabilities for the critical contingencies ranking. And for wide range systems, where the voltage amplitudes may not be a major problem, and the DC load flow provides appropriate accuracy in terms of active power flows and ignoring the reactive power flow and transmission line losses.

IV. POWERWORLD SIMULATION FOR MULTIPLE CONTINGENCY N-K SCENARIO

In this section, I will extend the study of contingency analysis to cover more outages or failure events at the same time, and I will continue the investigation for the bus outages that could happen because of any such missile attack on the critical substation or for any natural reasons such as earthquakes, hurricanes, floods, etc.

A. *Multiple Contingency Ranking*

After performing the contingency analysis for each bus outage, which represents simply the multiple contingency scenarios, where each bus connects at least two elements in the system, I proceed the critical bus ranking using new developed performance index combines the active and reactive performance indices represent early in section (II). As shown in equation (7) below:

$$PI_{Complex} = \sqrt{PI_{MW}{}^2 + PI_{QV}{}^2} \tag{7}$$

Where:

PI_{MW} Active power performance index

PI_{QV} Voltage performance index

As result for the different effects of the bus outages when

TABLE 5. Active power performance index using N-K contingency

Contingency	**Line Limit**	**Bus2**	**Bus4**	**Bus5**	**Bus6**
Line No.	**MVA**	**MVA**	**MVA**	**MVA**	**MVA**
Line 1	200			242.39	
Line 2	100	311.30			
Line 3	100			114.53	
Line 4	100			137.23	
Line 6	50	119.30			
Line 7	100	201.50			
Line 8	50			58.37	51.39
Line 15	50			59.81	56.27
Line 16	20		25.82	33.93	
Line 17	20			28.16	42.91
Line 18	20		32.30	24.08	
Line 20	20		23.82		23.03
Active Power PI		**19.44**	**5.69**	**13.77**	**8.25**

we monitor the transmission line power flow and calculate the active power performance index in table (5), we conclude that the most severe outage occurred when the Bus2 substation got out of service with PI_{MW} = 19.44 due to three lines overloads, then Bus5 outage with PI_{MW} = 13.77 due to eight lines overloads.

But consider the bus voltages for the remaining buses and calculate the voltage performance index in table (6), it's observed that the severe outage occurred when the Bus6 substation got out of service with PI_{QV} = 2.84 due to two buses goes under voltage limits, then Bus5 with PI_{QV} = 2.60 due to four buses goes under voltage limits.

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International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
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TABLE 6. Voltage performance index using N-K contingency

Contingency	Line Limit	Bus2	Bus4	Bus5	Bus6
Bus No.	MVA	MVA	MVA	MVA	MVA
Bus 5	0.92	0.89			
Bus 6	0.96			0.90	
Bus 12	0.96			0.88	0.80
Bus 13	0.95			0.88	0.83
Bus 14	0.93			0.89	
	Voltage PI	0.26	0.00	2.60	2.84

Finally, to determine the most critical and sensitive bus, by applying the complex performance index represented in equation (7) as shown below

As a conclusion form table (7), we notice that the most severe outage affecting the system drastically and leading to a blackout is the outage of Bus2 with $PI_{Complex}$ = 19.44 due to three lines overloads, one bus going under voltage limits, and other seven elements going out-of-service as shown in figure (2)

Also, we can notice the performance index ranking procedure is totally independent to determine the critical cases regardless the number of disturbed element nor exceed the limits

TABLE 7. Complex performance index using Multiple contingency Analysis

Contingency Summary	Bus2	Bus4	Bus5	Bus6
Total Violations	4	3	12	6
Total Disturbed Elements	7	7	6	7
Complex performance index	19.44	5.69	13.77	8.26
Contingency Ranking	**1**	**4**	**2**	**3**

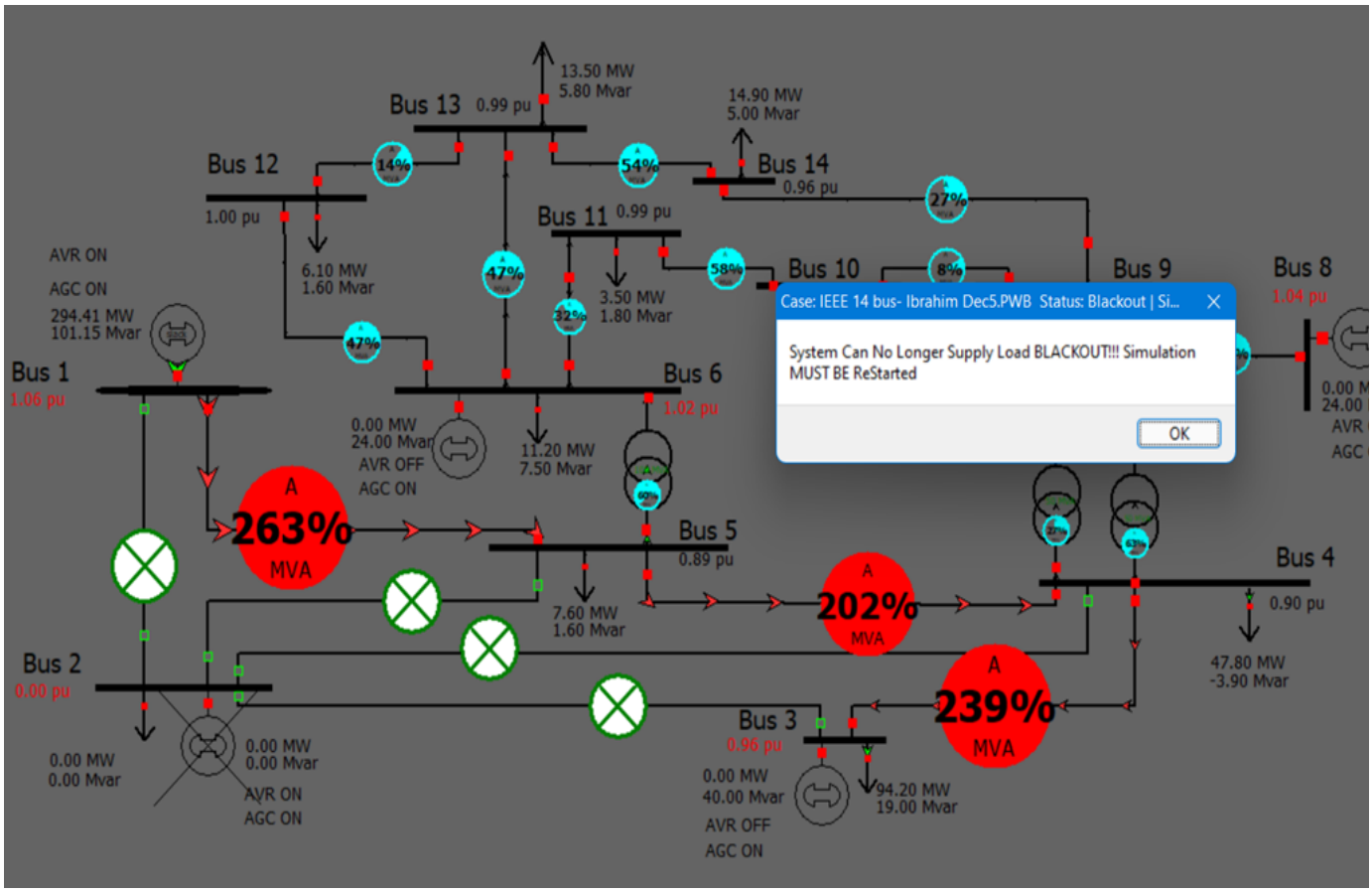


FIGURE 3. System blackout due to multiple contingencies occurred on the critical bus2

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International Journal of Engineering Research & Technology (IJERT)
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B. Critical Bus PV Curve

In this section, I will investigate the system stability using PV Curves for the critical buses and the operating zone effects for each of the buses if we apply a single contingency analysis, Figure (4) illustrates the PV curves for the most critical buses obtained before as shown below

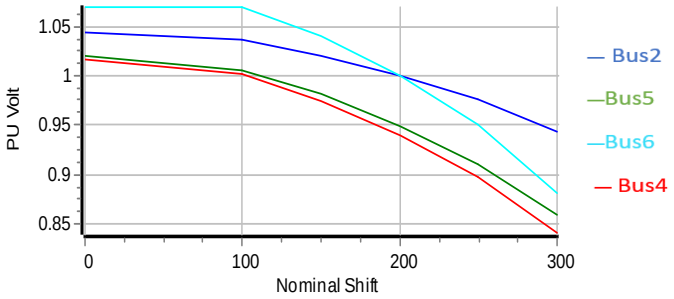


FIGURE 4. PV curves for the critical buses

Following the implementation of the PV curve analysis for the critical buses, I chose the top three critical line contingencies and notified the drastic shrinking in the operating zone, as well as the significant reduction in maximum power capacity.

Such a line 10 outage will reduce the voltage tolerance to less than 2% at 100MW power transfer. After that, the bus voltage will collapse and lead to a system blackout, as shown in figure (5), where the voltage tolerance was around 10% and the maximum power transfer exceed 300 MW.

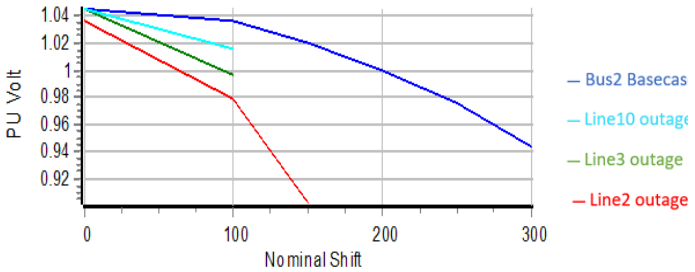


FIGURE 5. PV curves for Bus2 and the highest PI contingences

Similarly, we can generate the PV curve for all critical buses and conclude the maximum active power transfer during the top-ranked single contingency using the different types of performance indices.

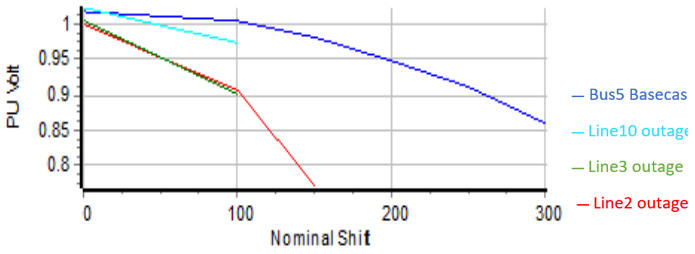


FIGURE 6. PV curves for Bus5 and the highest PI contingences

Again, for Bus5, the line10 outage will reduce the voltage tolerance to less than 2% at 100MW power transfer, as shown above in figure (6)

However, for Bus6, line 2 outage will reduce the voltage tolerance to less than 8% at 100MW power transfer, as shown in figure (7)

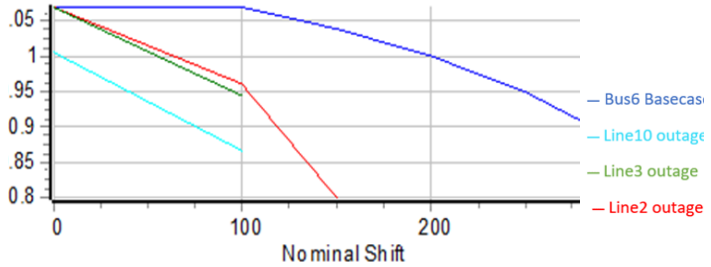


FIGURE 7. PV curves for Bus6 and the highest PI contingences

Finally, for Bus4, line 10 outage will reduce the voltage tolerance to less than 6% at 100MW power transfer, as shown in figure (8)

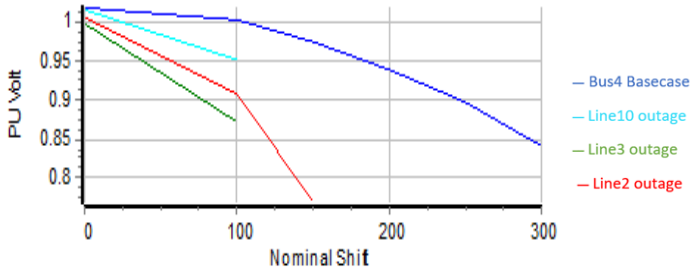


FIGURE 8. PV curves for Bus4 and the highest PI contingences

C. Critical Bus QV Curve

In this section, I will investigate the system stability using QV Curves for the critical buses and the operating zone effects for each of the buses if we apply a single contingency analysis, Figure (8) illustrates the QV curves for the most critical buses we obtained in previous section for multiple contingency ranking.

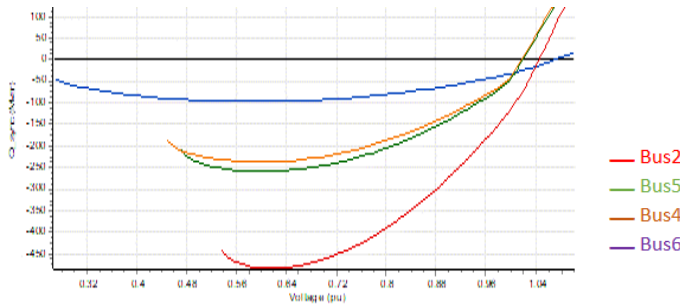


FIGURE 9. QV curves for the critical buses

Following the implementation of the QV curve analysis for the critical buses, I chose the top four critical line contingencies and notified the voltage tolerance in the

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International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 11 Issue 02, February-2022

operating zone, as well as the sensitivity factor (dQ/dV), which notifies us of the allowable modification in the reactive power injection to the available voltage variance, as represented below Table (8).

As shown in figure (10), Form Bus2 point of view, a line 1 outage will reduce the voltage tolerance to less than 16% and increase the reactive power injection to keep the system under stable condition with very little dQ/dV = 1%. And if the required injection increases further, the bus voltage will collapse and lead to a system blackout.

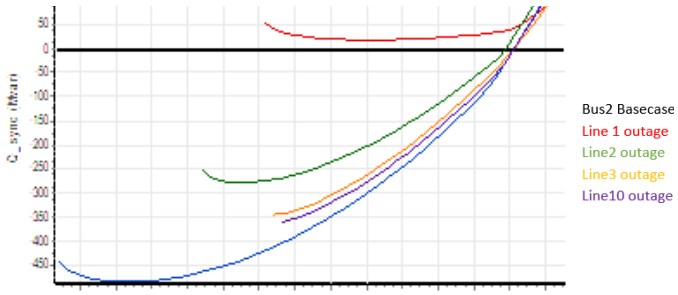


FIGURE 10. QV curves for Bus2 and the highest PI contingences

Similarly, we can generate the QV for Bus4, a line 1 outage will reduce the voltage tolerance to less than 17% and increase the reactive power injection to keep the system under stable condition with very little dQ/dV = 2%.

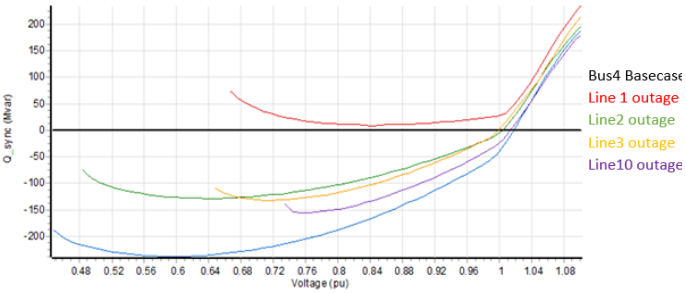


FIGURE 11. QV curves for Bus4 and the highest PI contingences

Also, For Bus5, a line 1 outage will reduce the voltage tolerance to less than 17% and increase the reactive power injection to keep the system under stable condition with very little dQ/dV = 2%.

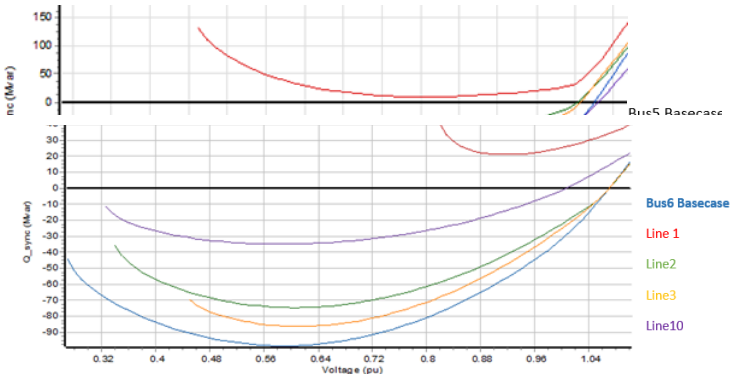


FIGURE 13. QV curves for Bus6 and the highest PI contingences

Finally, For Bus6, a line 1 outage will reduce the voltage tolerance to less than 14% and increase the reactive power injection to keep the system under stable condition with very little dQ/dV = 1%.

V. CONCLUSION

This paper is representing the different contingency ranking and screening method using linear sensitivity factors GFD and LODF for faster response of the system operators to estimate the post- contingency active power flow for all the system transmission lines. And the major of the study depends on utilizing the built-in function for the Powerworld simulator to investigate the N-1 contingency scenario and ranking the most critical lines and buses using different formulas of performance indices, and finally evaluate the critical buses voltage stability using PV and QV curves and assess the stability under the highest PI contingences.

TABLE 8. Voltage stability and critical buses sensitivity for the highest PI contingencies

	Case Name	V_{min}	Q_{min}	Voltage Tolerance	dQ/dV
Bus 2	**Bus2 Basecase**	**0.615**	**-441.12**	**41%**	**8%**
	Line 1 outage	0.885	60.93	16%	1%
	Line2 outage	0.7358	-229.32	29%	7%
	Line3 outage	0.775	-302.49	25%	9%
	Line10 outage	0.785	-315.88	25%	10%
Bus 4	**Bus4 Basecase**	**0.5977**	**-238.07**	**41%**	**4%**
	Line 1 outage	0.8477	9.84	17%	2%
	Line2 outage	0.6445	-128.18	36%	3%
	Line3 outage	0.7184	-131.85	28%	3%
	Line10 outage	0.764	-155.77	24%	5%
Bus 5	**Bus5 Basecase**	**0.5995**	**-259.44**	**41%**	**4%**
	Line 1 outage	0.8495	9.73	17%	2%
	Line2 outage	0.6318	-123.99	37%	2%
	Line3 outage	0.7256	-158.65	28%	4%
	Line10 outage	0.7443	-192.74	27%	5%
Bus 6	**Bus6 Basecase**	**0.59**	**-85.43**	**45%**	**1%**
	Line 1 outage	0.92	33.49	14%	1%
	Line2 outage	0.6	-51.02	44%	1%
	Line3 outage	0.61	-64.49	43%	1%
	Line10 outage	0.5873	-10.89	42%	1%

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International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 11 Issue 02, February-2022

VI. ACKNOWLEDGMENT

The authors gratefully would like to acknowledge the support and help that have been provided by King Abdulaziz University.

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International Journal of Engineering Research & Technology (IJERT)
ISSN: 2278-0181
Vol. 11 Issue 02, February-2022

Line	Location		Base-Case	Limit	Line 10 Bus5-6			Line 1 Bus1-2			Line 3 Bus2-3			Line 2 Bus1-5		
Number	From	To	MVA	MVA	MVA	PI	%	MVA	PI	%	MVA	PI	%	MVA	PI	%
1	Bus 1	Bus 2	147.9	200	150.4	0.57	75%	0	-	0%	133.30	0.44	67%	219.00	1.20	110%
2	Bus 1	Bus 5	71.1	100	68.6	0.47	69%	219	4.80	219%	85.70	0.73	86%	-	-	0%
3	Bus 2	Bus 3	70.1	100	72.2	0.52	72%	45.1	0.20	45%	-	-	0%	82.10	0.67	82%
4	Bus 2	Bus 4	55.2	100	59.7	0.36	60%	3	0.00	3%	87.10	0.76	87%	80.40	0.65	80%
5	Bus 2	Bus 5	40.9	100	36.8	0.14	37%	29.8	0.09	30%	64.50	0.42	65%	74.90	0.56	75%
6	Bus 3	Bus 4	24.1	50	22	0.19	44%	49.1	0.96	98%	94.20	3.55	188%	12.10	0.06	24%
7	Bus 4	Bus 5	62.3	100	97.8	0.96	98%	135.4	1.83	135%	98.40	0.97	98%	27.20	0.07	27%
8	Bus 4	Bus 7	29	50	55.7	1.24	111%	26.3	0.28	53%	27.70	0.31	55%	30.30	0.37	61%
9	Bus 4	Bus 9	16.6	50	32	0.41	64%	15.1	0.09	30%	15.90	0.10	32%	17.40	0.12	35%
10	Bus 5	Bus 6	42.1	100	0	-	0%	46.2	0.21	46%	44.10	0.19	44%	40.10	0.16	40%
11	Bus 6	Bus 11	6.3	50	19	0.14	38%	8.8	0.03	18%	7.50	0.02	15%	5.10	0.01	10%
12	Bus 6	Bus 12	7.5	20	3.8	0.04	19%	7.9	0.16	40%	7.70	0.15	39%	7.40	0.14	37%
13	Bus 6	Bus 13	17	50	4	0.01	8%	18.3	0.13	37%	17.70	0.13	35%	16.40	0.11	33%
14	Bus 7	Bus 8	0	50	0	-	0%	0	-	0%	-	-	0%	-	-	0%
15	Bus 7	Bus 9	29	50	55.7	1.24	111%	26.3	0.28	53%	27.70	0.31	55%	30.30	0.37	61%
16	Bus 9	Bus 10	6.2	20	31.5	2.48	158%	3.7	0.03	19%	5.00	0.06	25%	7.40	0.14	37%
17	Bus 9	Bus 14	9.9	20	26.7	1.78	134%	8.3	0.17	42%	9.10	0.21	46%	10.70	0.29	54%
18	Bus 10	Bus 11	2.8	20	22.5	1.27	113%	5.3	0.07	27%	4.00	0.04	20%	1.60	0.01	8%
19	Bus 12	Bus 13	1.4	20	2.3	0.01	12%	1.8	0.01	9%	1.60	0.01	8%	1.30	0.00	7%
20	Bus 13	Bus 14	5	20	11.8	0.35	59%	6.6	0.11	33%	5.80	0.08	29%	4.20	0.04	21%
					PI	12.17		PI	9.46		PI	8.48		PI	4.96	

APPENDIX 1. Performance index for N-1 contingency Analysis using DC load flow

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International Journal of Engineering Research & Technology (IJERT)
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Line Number	Location From	Location To	Base-Case MVA	Limit MVA	L10 Bus5-6 MVA	L10 Bus5-6 PI	L10 Bus5-6 %	L3 Bus2-3 MVA	L3 Bus2-3 PI	L3 Bus2-3 %	L13 Bus6-13 MVA	L13 Bus6-13 PI	L13 Bus6-13 %	L2 Bus1-5 MVA	L2 Bus1-5 PI	L2 Bus1-5 %
1	Bus 1	Bus 2	158.2	200	162.9	0.66	81%	149.5	0.56	75%	159.10	0.63	80%	241.00	1.45	121%
2	Bus 1	Bus 5	75.6	100	74	0.55	74%	95.4	0.91	95%	75.70	0.57	76%	-	-	0%
3	Bus 2	Bus 3	73.3	100	76.1	0.58	76%	0	-	0%	73.70	0.54	74%	87.20	0.76	87%
4	Bus 2	Bus 4	56.2	100	61.6	0.38	62%	93.7	0.88	94%	56.70	0.32	57%	83.40	0.70	83%
5	Bus 2	Bus 5	41.5	100	37.7	0.14	38%	69.2	0.48	69%	41.50	0.17	42%	77.90	0.61	78%
6	Bus 3	Bus 4	23.7	50	21.4	0.18	43%	96.5	3.72	193%	23.60	0.22	47%	12.40	0.06	25%
7	Bus 4	Bus 5	63.2	100	100	1.00	100%	102.4	1.05	102%	65.40	0.43	65%	28.50	0.08	29%
8	Bus 4	Bus 7	29.7	50	57.6	1.33	115%	30	0.36	60%	31.40	0.39	63%	32.10	0.41	64%
9	Bus 4	Bus 9	16.1	50	32.9	0.43	66%	15.1	0.09	30%	17.30	0.12	35%	16.80	0.11	34%
10	Bus 5	Bus 6	45.8	100	0	-	0%	47.7	0.23	48%	43.20	0.19	43%	42.50	0.18	43%
11	Bus 6	Bus 11	8.2	50	20.2	0.16	40%	10.5	0.04	21%	11.40	0.05	23%	8.10	0.03	16%
12	Bus 6	Bus 12	8.2	20	5.4	0.07	27%	8.5	0.18	43%	20.60	1.06	103%	8.20	0.17	41%
13	Bus 6	Bus 13	19.2	50	8.8	0.03	18%	20.4	0.17	41%	-	-	0%	19.00	0.14	38%
14	Bus 7	Bus 8	17.2	50	21.9	0.19	44%	22.5	0.20	45%	19.60	0.15	39%	20.50	0.17	41%
15	Bus 7	Bus 9	28.7	50	58.3	1.36	117%	26.8	0.29	54%	31.70	0.40	63%	29.70	0.35	59%
16	Bus 9	Bus 10	6.7	20	33.5	2.81	168%	4.5	0.05	23%	4.40	0.05	22%	6.90	0.12	35%
17	Bus 9	Bus 14	10.1	20	27.8	1.93	139%	8.6	0.18	43%	18.10	0.82	91%	10.40	0.27	52%
18	Bus 10	Bus 11	4.1	20	23.8	1.42	119%	6.3	0.10	32%	7.30	0.13	37%	4.20	0.04	21%
19	Bus 12	Bus 13	1.8	20	2.4	0.01	12%	2.1	0.01	11%	13.60	0.46	68%	1.80	0.01	9%
20	Bus 13	Bus 14	5.9	20	12	0.36	60%	7.4	0.14	37%	2.50	0.02	13%	5.70	0.08	29%
					PI	**13.60**		**PI**	**9.64**		**PI**	**6.74**		**PI**	**5.75**	

APPENDIX 2. Performance index for N-1 contingency Analysis using AC load flow

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